The Origin of G_2 Holonomy: Vacuum Stabilization via Sedenion Zero-Divisor Exclusion

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Abstract

We propose a cosmological framework wherein the pre-geometric vacuum is identified with the full hierarchy of Cayley-Dickson algebras ($\mathbb{A}_n$). We demonstrate that the "Big Bang" corresponds to a topological filtration event driven by the distinct algebraic properties of Time and Space. While the metric geometry of space collapses in higher dimensions due to the proliferation of zero divisors ($xy = 0$), the temporal evolution remains unitary due to the universal property of *Power Associativity* ($x^n x^m = x^{n+m}$). We identify the physical vacuum as the stable residue of this filtration. Utilizing recent results by Reggiani (2024), we prove that the manifold of excluded Sedenion zero divisors $\mathcal{Z}(\mathbb{S})$ is isometric to the exceptional Lie group G_2. Consequently, the observed compactification of M-theory on a manifold of G_2 holonomy is derived as the boundary condition of the algebraic void left by the expulsion of the zero-divisor swarm.

1 Introduction: The Algebraic Ecology of the Vacuum

Standard M-theory compactifications rely on manifolds of G_2 holonomy to preserve $\mathcal{N} = 1$ supersymmetry in four dimensions [1]. However, the selection principle for this specific holonomy group remains an open question. We propose that G_2 geometry is not an arbitrary choice, but a necessary boundary condition imposed by the algebraic stability of the pre-geometric vacuum.

We postulate that the initial cosmological state is a **Plenum** containing the full tower of Cayley-Dickson algebras $\mathbb{A}_n$ (Real, Complex, Quaternion, Octonion, Sedenion, Trigintaduonion, ...). The dynamics of cosmogenesis are driven by the competition between two fundamental imperatives:

1. **Temporal Persistence:** The requirement for unitary time evolution $U(t)$.
2. **Spatial Observability:** The requirement for a non-degenerate metric norm $\|xy\| = \|x\|\|y\|$.

2 The Power-Associative Lifeline: The Persistence of Time

A critical distinction exists between the algebraic requirements for time versus space.

2.1 Universal Unitary Evolution

Time evolution is generated by the exponentiation of a Hamiltonian operator H. A fundamental property preserved throughout the entire Cayley-Dickson construction (even for $n \geq 4$) is **Power Associativity**:

$$x^n \cdot x^m = x^{n+m}, \quad \forall x \in \mathbb{A}_n \tag{1}$$

This property ensures that the one-parameter group of time evolution operators is well-defined even in the high-dimensional bulk algebras (e.g., Trigintaduonions $\mathbb{T}$):

$$U(t) = e^{-iHt} = \sum_{k=0}^{\infty} \frac{(-iHt)^k}{k!} \tag{2}$$

Because powers associate, $U(t_1)U(t_2) = U(t_1 + t_2)$. Therefore, **Time is fundamental**. The causal clock ticks consistently regardless of the dimension or non-associativity of the substrate.

2.2 The Spatial Catastrophe

In contrast, spatial geometry relies on the existence of a division algebra to define a consistent metric and interaction vertex. For $n \geq 4$ (Sedenions $\mathbb{S}$ and beyond), the algebras lose the division property and admit **Zero Divisors**:

$$\exists u, v \in \mathbb{A}_n, u \neq 0, v \neq 0 \text{ such that } u \cdot v = 0 \tag{3}$$

Physically, a zero divisor represents a **Vacuum Crash**. If two non-zero states interact to produce zero, information is destroyed locally. The spatial metric becomes singular, and the norm property fails ($\|uv\| \neq \|u\|\|v\|$). Thus, while Time persists in the bulk, Space is dynamically unstable.

3 The Reggiani Bridge: Deriving G_2 from Failure

To satisfy the Axiom of Observability for spatial interactions, the vacuum must excise the zero-divisor regions. We focus on the first critical failure at $n = 4$ (Sedenions).

3.1 The Manifold of Zero Divisors

Let $\mathcal{Z}(\mathbb{S}) \subset \mathbb{S} \times \mathbb{S}$ be the manifold of normalized zero divisors in the Sedenion algebra. The physical vacuum $\mathcal{M}_{phys}$ is the complement:

$$\mathcal{M}_{phys} \cong \mathbb{S} \setminus \mathcal{Z}(\mathbb{S}) \tag{4}$$

The topology of the observable universe is determined by the boundary of this excluded hole.

3.2 Theorem: The Isometry to G_2

We utilize the rigorous result established by Reggiani (2024):

> **Theorem 1 (Reggiani):** The manifold of zero divisors $\mathcal{Z}(\mathbb{S})$ is isometric to the compact exceptional Lie group G_2 equipped with a naturally reductive left-invariant metric.

$$\mathcal{Z}(\mathbb{S}) \cong G_2 \tag{5}$$

This establishes a direct link between algebraic failure and geometric structure. The G_2 manifold of M-theory is physically identified as the **Topological Scar** formed by the removal of Sedenionic zero divisors from the vacuum. The 7 compact dimensions are necessary to wrap around this 14-dimensional defect.

4 Cosmological Selection: Inflation as Filtration

The *Big Bang* is identified as the phase transition $\mathbb{T} \to \mathbb{S} \to \mathbb{O}$.

4.1 Inflationary Dilution

The density of zero divisors in the pre-geometric bulk scales exponentially with dimension. To stabilize the Octonionic sector ($\mathbb{O}$), the vacuum initiates **Cosmic Inflation**.

$$a(t) \propto e^{Ht} \tag{6}$$

This exponential expansion is not a creation event but a **Dilution Mechanism**. It drives the local density of zero-divisor regions $\rho_{\mathcal{Z}}$ to zero, pushing the algebraic singularities beyond the causal horizon. Inflation halts when the vacuum crystallizes into the division algebra $\mathbb{O}$, where local spatial interactions become reversible ($xy = 0 \implies x = 0$).

4.2 Cosmic Voids as Fossils

Regions of the vacuum that failed to clear the zero-divisor condition during the phase transition persist as **Cosmic Supervoids**.

$$\text{Void} \equiv \text{Region where } \langle \mathcal{Z}(\mathbb{S}) \rangle \neq 0 \tag{7}$$

We predict that these voids are not empty but **Causally Disconnected**. They should exhibit anomalous lensing and photon propagation properties (opacity) due to the failure of the metric connection in the zero-divisor limit.

5 Conclusion

We have derived the G_2 holonomy of M-theory from the algebraic properties of the Sedenions. The universe exists in a delicate balance: it rides the **Power-Associative Lifeline** of time while constantly filtering out the zero-divisor instabilities of space. The observed physics of the Standard Model is the vibration of the associative strings trapped on the surface of this algebraic exclusion zone.